Lecture 2: Asymptotic Analysis, Recurrences

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Sep 3, 2026
601.433/633 Introduction to Algorithms
Slides from Mike Dinitz, modified for our section

Administrative

- Schedule change! See course website
- Oral Homeworks:The oral-presentation homeworks will be done in groups of three. Each of these assignments will consist of three problems. The members of your group will work together to solve the problems (so, unlike written homeworks, here collaboration isrequired). You will then present your solutions, as a group, to one of the instructors or TAs.
- Presentations will be given in 1-hour time slots (there will be an electronic sign-up sheet reachable from the course home page). At the presentation, each member of the group will spend 15 minutes presenting one of the problems. The instructor (or TA) will decide who presents which problem, but when one member is presenting, other members are allowed to chime in too. In the end, the three presentations together will determine the grade for the group.
- If you are nervous about your presentation, you may in addition hand in a written sketch of your solution as well. We will then take this writeup into consideration in determining your grade on the assignment.

Today

Should be review, some might be new.
See math background in CLRS

Asymptotics: $\boldsymbol{O(\cdot)}$, $\boldsymbol{\Omega(\cdot)}$, and $\boldsymbol{\Theta(\cdot)}$ notation.

- Should know from Data Structures / MFCS. We'll be a bit more formal.
- Intuitively: hide constants and lower order terms, since we only care what happens "at scale" (asymptotically)

Recurrences: How to solve recurrence relations.

- Should know from MFCS / Discrete Math.

Asymptotic Notation

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Abuse notation: "$g(n)$ is $O(f(n))$" or $g(n) = O(f(n))$.

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Examples:

- $2n^2 + 27 = O(n^2)$: set $n_0 = 6$ and $c = 3$
- $2n^2 + 27 = O(n^3)$: same values, or $n_0 = 4$ and $c = 1$
- $n^3 + 2000n^2 + 2000n = O(n^3)$: set $n_0 = 10000$ and $c = 2$

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About *functions* not algorithms!
Expresses an *upper* bound

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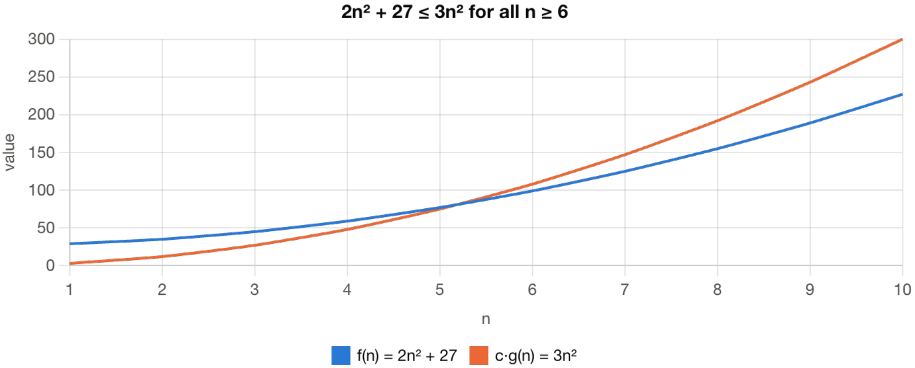

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Set $c = 3$. Suppose $2n^2 + 27 > cn^2 = 3n^2$

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$\implies n^2 < 27 \implies n < 6$

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Set $n_0 = 6$. Then $2n^2 + 27 \leq cn^2$ for all $n > n_0$. $\square$

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Set $n_0 = 6$. Then $2n^2 + 27 \leq cn^2$ for all $n > n_0$. $\square$

Many other ways to prove this!

$\Omega(\cdot)$

Counterpart to $O(\cdot)$: *lower* bound rather than upper bound.

Definition

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Examples:

- $2n^2 + 27 = \Omega(n^2)$: set $n_0 = 1$ and $c = 1$
- $2n^2 + 27 = \Omega(n)$: set $n_0 = 1$ and $c = 1$
- $\frac{1}{100}n^3 - 1000n^2 = \Omega(n^3)$: set $n_0 = 1000000$ and $c = 1/1000$

$\Theta(\cdot)$

Combination of $O(\cdot)$ and $\Omega(\cdot)$.

Definition

$g(n) \in \Theta(f(n))$ if $g(n) \in O(f(n))$ and $g(n) \in \Omega(f(n))$.

Note: constants n_0, c can be different in the proofs for $O(f(n))$ and $\Omega(f(n))$

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Equivalent:

Definition

$g(n) \in \Theta(f(n))$ if there are constants $c_1, c_2, n_0 > 0$ such that $c_1 f(n) \leq g(n) \leq c_2 f(n)$ for all $n > n_0$.

Both lower bound and upper bound, so asymptotic equality.

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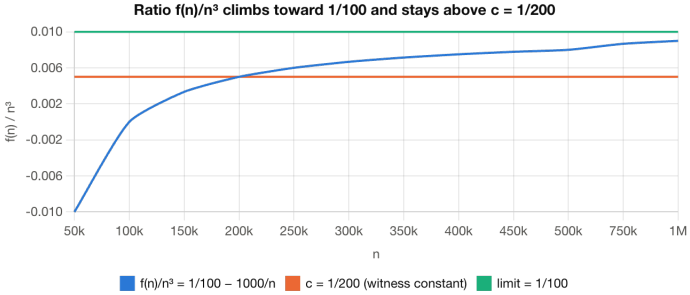
Example: $\frac{1}{100}n^3 - 1000n^2 = \Omega(n^3)$: set $n_0 = 1000000$ and $c = 1/1000$



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Little notation

Strict versions of O and Ω:

Definition

$g(n) \in o(f(n))$ if for every constant $c > 0$ there exists a constant $n_0 > 0$ such that $g(n) < c \cdot f(n)$ for all $n > n_0$.

Definition

$g(n) \in \omega(f(n))$ if for every constant $c > 0$ there exists a constant $n_0 > 0$ such that $g(n) > c \cdot f(n)$ for all $n > n_0$.

Examples:

- $2n^2 + 27 = o(n^2 \log n)$
- $2n^2 + 27 = \omega(n)$

Recurrence Relations

Sorting

Many algorithms recursive so running time naturally a recurrence relation (Karatsuba, Strassen).

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  - Running time: Takes $O(n)$ time to find smallest unsorted element, decreases remaining unsorted by 1.
    $\implies T(n) = T(n-1) + cn$

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Also need base case. For algorithms, constant size input takes constant time.
$\implies T(n) \le c$ for all $n \le n_0$, for some constants $n_0, c > 0$.

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$$\begin{aligned} T(n) = 3T(n/3) + n \leq 3(n/3)\log_3(3n/3) + n &= n\log_3(n) + n \\ = n(\log_3(n) + \log_3 3) &= n\log_3(3n). \end{aligned}$$

Unrolling

Example: selection sort. $T(n) = T(n-1) + cn$

Idea: "unroll" the recurrence.

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At least $n/2$ terms which are at least $cn/2 \implies T(n) \ge c\frac{n^2}{4} = \Omega(n^2)$

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$\implies T(n) = \Theta(n^2)$.

Recursion Tree: Mergesort

Generalizes unrolling: draw out full tree of “recursive calls”.

Mergesort: $T(n) = 2T(n/2) + cn$.

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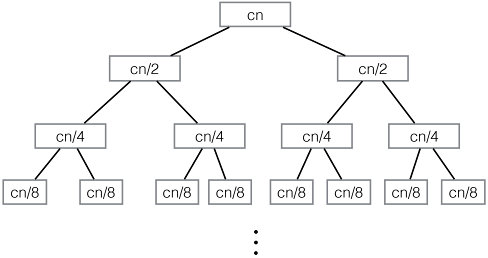

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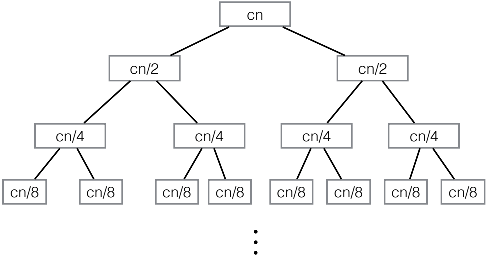

levels:

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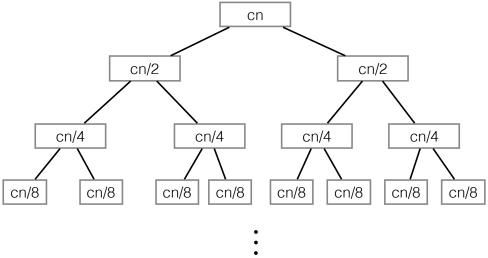

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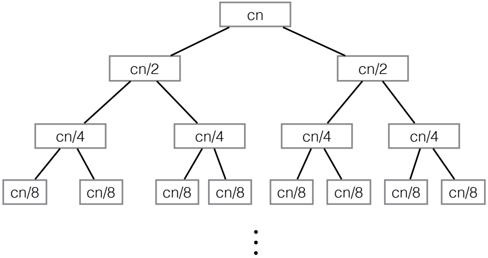

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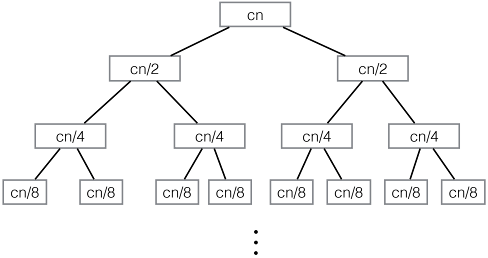

levels: $\log_2 n$

Contribution of level i: $2^{i-1} cn/2^{i-1} = cn$

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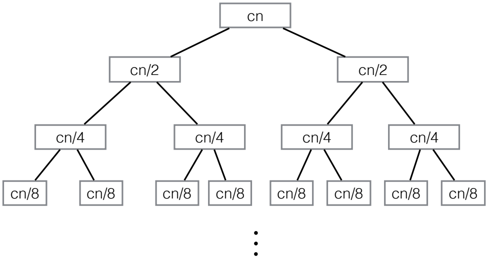

\# levels: $\log_2 n$

Contribution of level i: $2^{i-1} cn/2^{i-1} = cn$

$\implies T(n) = \Theta(n \log n)$

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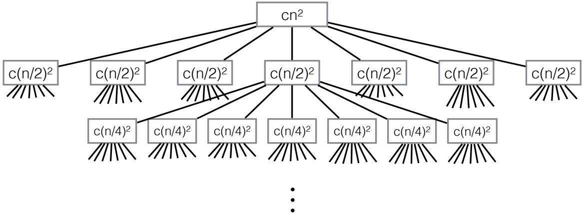

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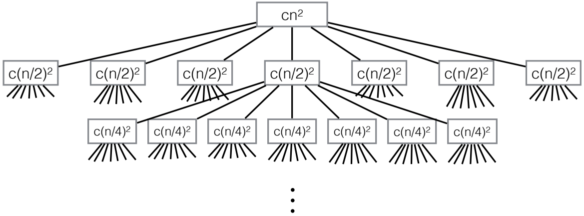

Level i: $7^{i-1} c(n/2^{i-1})^2 = (7/4)^{i-1} cn^2$

Recursion Tree: Strassen

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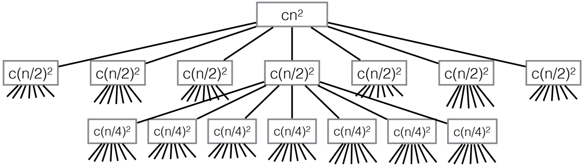

$$\vdots$$

Level i: $7^{i-1}c(n/2^{i-1})^2 = (7/4)^{i-1}cn^2$

Total:

$$T(n) = \sum_{i=1}^{\log n+1} \left(\frac{7}{4}\right)^{i-1} cn^2 = cn^2 \sum_{i=1}^{\log n+1} \left(\frac{7}{4}\right)^{i-1}$$

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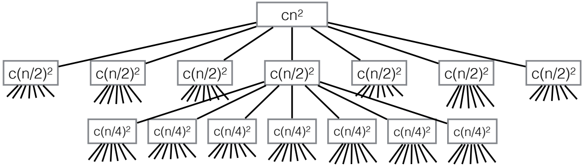

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$$\implies T(n) = O(n^2(7/4)^{\log_2 n}) = O(n^2 2^{\log_2(7/4)\cdot\log_2 n}) = O(n^2 n^{\log_2(7/4)}) = O(n^2 n^{\log_2 7-2})$$

$$= O(n^{\log_2 7})$$

Master Theorem

$$T(n) = aT(n/b) + cn^k \qquad\qquad T(1) = c$$

a, b, c, k constants with $a \geq 1$, $b > 1$, $c > 0$, and $k \geq 0$

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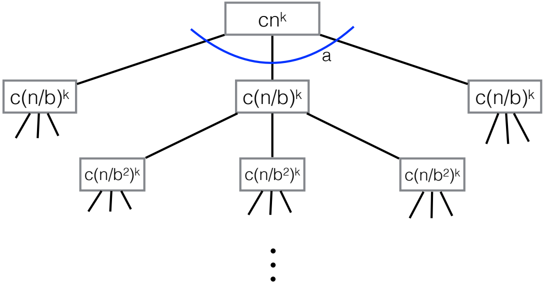

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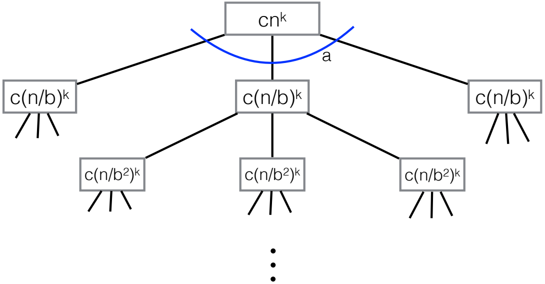

levels: $\log_b n + 1$

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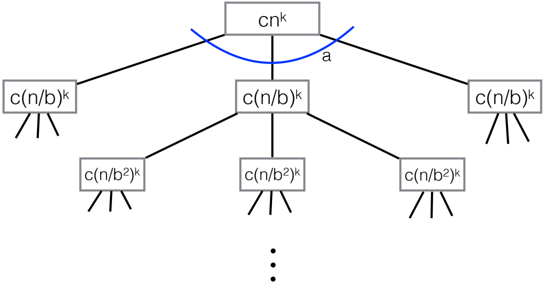

levels: $\log_b n + 1$

Level i: $a^{i-1}c(n/b^{i-1})^k = cn^k(a/b^k)^{i-1}$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

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- Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

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  $\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \frac{1}{1-\alpha}$.
  $\implies T(n) = O(n^k)$

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  $T(n) \geq cn^k \implies T(n) = \Omega(n^k)$

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- Case 3: $\alpha > 1$. Dominated by bottom level

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  $$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} = \alpha^{\log_b n} \sum_{i=1}^{\log_b n+1} \left(\frac{1}{\alpha}\right)^{i-1} \le \alpha^{\log_b n} \frac{1}{1-(1/\alpha)}$$

  $$= O(\alpha^{\log_b n})$$

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  $$= O(\alpha^{\log_b n})$$

  $$\implies T(n) = \Theta(n^k \alpha^{\log_b n}) = \Theta(n^k (a/b^k)^{\log_b n}) = \Theta(a^{\log_b n})$$

  $$= \Theta(n^{\log_b a})$$

Master Theorem III

Theorem ("Master Theorem")

The recurrence

$$T(n) = aT(n/b) + cn^k \qquad T(1) = c$$

where a, b, c, *and* k *are constants with* $a \geq 1$, $b > 1$, $c > 0$, *and* $k \geq 0$, *is equal to*

$$T(n) = \Theta(n^k) \text{ if } a < b^k,$$
$$T(n) = \Theta(n^k \log n) \text{ if } a = b^k,$$
$$T(n) = \Theta(n^{\log_b a}) \text{ if } a > b^k.$$